Suggested problems: 1-10

Problem 1: Suppose that we toss a coin six times. What is the cardinality of the sample space consisting of the outcomes of the six tosses?

Problem 2: A standard combination lock has a dial with tick marks for 40 numbers from 0 to 39. The combination consists of a sequence of three numbers that must be dialed in the correct order to open the lock. What is the cardinality of the sample space consisting of all possible combinations?

Problem 3: Consider an experiment in which a card is selected and removed from a deck of n different cards, a second card is then selected and removed from the remaining $n-1$ cards, and finally a third card is selected from the remaining $n-2$ cards. Each outcome consists of the three cards in the order selected. What is the cardinality of the sample space consisting of all possible selections of three cards?

Problem 4: Suppose that a club consists of 25 members and that a president and a secretary are to be chosen from the membership. Determine the total number of ways in which these two positions can be filled.

Problem 5: (The Birthday Problem.) What is the smallest number k such that, in a room of k people, the probability that at least two people share a birthday is greater than $1/2$?